

Python EDA + SQLite3 Assignment

Full Worked Solution — Code & Output

Dataset: EDA_SQLite_Employee_200.csv (200 employee records, with injected duplicates and missing values)

Note: No CSV file was attached to the assignment brief, so this solution generates a realistic synthetic dataset (EDA_SQLite_Employee_200.csv) matching the described shape — 200 employee records with the required columns, plus injected duplicate rows and missing values — and solves all 50 tasks and the bonus challenges against it.

Part 1: Load And Inspect Data

Code

```
section("PART 1: LOAD AND INSPECT DATA")

# 1-2. Import libraries (pandas, numpy, sqlite3) already imported above; read CSV
df = pd.read_csv("EDA_SQLite_Employee_200.csv")

# 3. head(), tail(), sample(10)
print("\n--- head() ---")
print(df.head())
print("\n--- tail() ---")
print(df.tail())
print("\n--- sample(10) ---")
print(df.sample(10, random_state=1))

# 4. shape, columns, dtypes
print("\n--- shape ---")
print(df.shape)
print("\n--- columns ---")
print(list(df.columns))
print("\n--- dtypes ---")
print(df.dtypes)

# 5. info(), describe(include='all')
print("\n--- info() ---")
df.info()
print("\n--- describe(include='all') ---")
print(df.describe(include="all"))

# 6. Set EmployeeID as index
df = df.set_index("EmployeeID")
print("\n--- After set_index('EmployeeID') ---")
print(df.head())
```

Output

```
--- head() ---
   EmployeeID  Name  Age  Department  City  Gender  Salary  ExperienceYears  PerformanceScore  J
0      1109  Dina Mostafa  33.0      Sales  Cairo  Male  13749.00              11.0              3.0  202
1      1170  Nesma Wilson  24.0      Sales  Tanta  Male  17588.45              0.0              5.0  202
2      1107  Yasmin Mostafa  33.0     Finance  Luxor  Male  8325.27              2.0              3.0  201
3      1108   Rana Fathy  40.0  Operations  Luxor  Female  8460.87             16.0              2.0  201
4      1052  David Abdallah  58.0          HR  Cairo  Male  5792.09              3.0              4.0  201

--- tail() ---
   EmployeeID  Name  Age  Department  City  Gender  Salary  ExperienceYears  PerformanceScore
210      1177  Mariam Ramzy  53.0          HR  Tanta  Female  13823.96              6.0              NaN
211      1138  Hassan Abdallah  28.0      Sales  Luxor  Female  10987.49              1.0              NaN
212      1073   James Fathy  28.0  Operations  Tanta  Female  10042.16             18.0              NaN
213      1141  Sherif Brown  48.0          IT  Tanta  Male  10004.41             12.0              5.0
214      1038   Nour Nassar  25.0          IT  Cairo  Male  11179.80              2.0              NaN

--- sample(10) ---
   EmployeeID  Name  Age  Department  City  Gender  Salary  ExperienceYears  PerformanceSc
108      1046  Ahmed Fahmy  41.0      Sales  Ismailia  Female  4000.00              14.0
109      1193   Dina Davis  37.0     Finance  Ismailia  Male  9151.45              5.0
110      1188  Robert Ramzy  35.0    Marketing  Tanta  Female  12411.98              7.0
```

```

107      1192      Laila Wilson  24.0  Operations      Giza      Male  9136.07      8.0
      NaN  2016-10-11
51      1079      Patricia Hassan  42.0      IT      Mansoura      Male  4000.00      16.0
      3.0  2021-05-09
33      1012      Patricia Fahmy      NaN      IT      Giza      Female  11617.93      5.0
      4.0  2011-05-04
59      1102      Aya Farouk  46.0      Sales  Alexandria  Female  7129.58      2.0
      5.0  2019-09-01
152     1164      Farida Davis  34.0      Sales      Mansoura  Female  8635.83      0.0
      5.0  2020-03-18
4      1052      David Abdallah  58.0      HR      Cairo      Male  5792.09      3.0
      4.0  2016-11-06
31     1182      Farida Smith  58.0      IT  Alexandria  Female  6493.80      17.0
      2.0  2024-12-25

--- shape ---
(215, 10)

--- columns ---
['EmployeeID', 'Name', 'Age', 'Department', 'City', 'Gender', 'Salary', 'ExperienceYears', 'PerformanceScore', 'JoinDate']

--- dtypes ---
EmployeeID      int64
Name            str
Age            float64
Department      str
City           str
Gender          str
Salary          float64
ExperienceYears float64
PerformanceScore float64
JoinDate        str
dtype: object

--- info() ---
<class 'pandas.DataFrame'>
RangeIndex: 215 entries, 0 to 214
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   EmployeeID            215 non-null   int64
1   Name                  215 non-null   str
2   Age                  205 non-null   float64
3   Department            215 non-null   str
4   City                  215 non-null   str
5   Gender                215 non-null   str
6   Salary                201 non-null   float64
7   ExperienceYears        204 non-null   float64
8   PerformanceScore       201 non-null   float64
9   JoinDate              215 non-null   str
dtypes: float64(4), int64(1), str(5)
memory usage: 16.9 KB

--- describe(include='all') ---

```

	EmployeeID	Name	Age	Department	City	Gender	Salary	ExperienceYears	PerformanceScore
count	215.000000	215	205.000000	215	215	215	201.000000	204.000000	2
unique	NaN	182	NaN	6	8	2	NaN	NaN	NaN
top	NaN	Yasmin Mostafa	NaN	IT	Aswan	Male	NaN	NaN	NaN
freq	NaN	3	NaN	54	31	119	NaN	NaN	NaN
mean	1101.023256	NaN	40.556098	NaN	NaN	NaN	9394.746517	9.588235	NaN
std	3.089552	NaN	11.094206	NaN	NaN	NaN	3090.371179	5.688729	NaN
min	1001.000000	NaN	22.000000	NaN	NaN	NaN	4000.000000	0.000000	NaN
25%	1050.500000	NaN	31.000000	NaN	NaN	NaN	7515.250000	5.000000	NaN
50%	1102.000000	NaN	40.000000	NaN	NaN	NaN	9190.120000	10.000000	NaN
75%	1150.500000	NaN	50.000000	NaN	NaN	NaN	11490.680000	14.000000	NaN
max	1200.000000	NaN	59.000000	NaN	NaN	NaN	18662.670000	19.000000	NaN

5.000000 NaN

--- After set_index('EmployeeID') ---

Date EmployeeID	Name	Age	Department	City	Gender	Salary	ExperienceYears	PerformanceScore	Join
1109 0-27	Dina Mostafa	33.0	Sales	Cairo	Male	13749.00	11.0	3.0	2023-1
1170 2-17	Nesma Wilson	24.0	Sales	Tanta	Male	17588.45	0.0	5.0	2023-0
1107 6-17	Yasmin Mostafa	33.0	Finance	Luxor	Male	8325.27	2.0	3.0	2016-0
1108 3-21	Rana Fathy	40.0	Operations	Luxor	Female	8460.87	16.0	2.0	2011-0
1052 1-06	David Abdallah	58.0	HR	Cairo	Male	5792.09	3.0	4.0	2016-1

Part 2: Data Cleaning

Code

```
section("PART 2: DATA CLEANING")

# 7. Missing values per column
print("\n--- Missing values per column (before cleaning) ---")
print(df.isnull().sum())

# 8. Fill missing Age with median
df["Age"] = df["Age"].fillna(df["Age"].median())

# 9. Fill missing Salary with mean
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())

# 10. Fill missing ExperienceYears with median
df["ExperienceYears"] = df["ExperienceYears"].fillna(df["ExperienceYears"].median())

# 11. Fill missing PerformanceScore with median
df["PerformanceScore"] = df["PerformanceScore"].fillna(df["PerformanceScore"].median())

# 12. Remove duplicate rows
before = len(df)
df = df.drop_duplicates()
print(f"\nRemoved {before - len(df)} duplicate rows (from {before} to {len(df)})")

# 13. Rename PerformanceScore -> Rating
df = df.rename(columns={"PerformanceScore": "Rating"})

# 14. Convert JoinDate to datetime
df["JoinDate"] = pd.to_datetime(df["JoinDate"])

# 15. Verify no missing values remain
print("\n--- Missing values per column (after cleaning) ---")
print(df.isnull().sum())
```

Output

```
--- Missing values per column (before cleaning) ---
Name      0
Age       10
Department 0
City       0
Gender     0
Salary    14
ExperienceYears  11
PerformanceScore  14
JoinDate   0
dtype: int64

Removed 15 duplicate rows (from 215 to 200)

--- Missing values per column (after cleaning) ---
Name      0
Age       0
Department 0
City       0
Gender     0
Salary     0
ExperienceYears  0
Rating     0
JoinDate   0
dtype: int64
```

Part 3: Data Manipulation

Code

```
section("PART 3: DATA MANIPULATION")

# 16. Bonus = Salary * 0.10
df["Bonus"] = df["Salary"] * 0.10

# 17. TotalSalary = Salary + Bonus
df["TotalSalary"] = df["Salary"] + df["Bonus"]

# 18. SeniorEmployee Yes/No if ExperienceYears >= 5
df["SeniorEmployee"] = np.where(df["ExperienceYears"] >= 5, "Yes", "No")

print("\n--- New columns preview ---")
print(df[["Salary", "Bonus", "TotalSalary", "ExperienceYears", "SeniorEmployee"]].head())

# 19. Filter IT employees
it_employees = df[df["Department"] == "IT"]
print(f"\n--- IT employees: {len(it_employees)} rows ---")
print(it_employees.head())

# 20. Filter Salary > 10000
high_salary = df[df["Salary"] > 10000]
print(f"\n--- Salary > 10000: {len(high_salary)} rows ---")
print(high_salary.head())

# 21. Filter employees from Cairo with Rating >= 4
cairo_top = df[(df["City"] == "Cairo") & (df["Rating"] >= 4)]
print(f"\n--- Cairo & Rating>=4: {len(cairo_top)} rows ---")
print(cairo_top.head())

# 22. Sort by Salary descending
by_salary_desc = df.sort_values("Salary", ascending=False)
print("\n--- Sorted by Salary desc (top 5) ---")
print(by_salary_desc.head())

# 23. Sort by JoinDate ascending
by_join_asc = df.sort_values("JoinDate", ascending=True)
print("\n--- Sorted by JoinDate asc (top 5) ---")
print(by_join_asc.head())

# 24. Top 10 highest salaries
top10_salary = df.sort_values("Salary", ascending=False).head(10)
print("\n--- Top 10 highest salaries ---")
print(top10_salary[["Name", "Department", "Salary"]])
```

Output

```
--- New columns preview ---
      Salary      Bonus  TotalSalary  ExperienceYears  SeniorEmployee
EmployeeID
1109      13749.00    1374.900    15123.900           11.0             Yes
1170      17588.45    1758.845    19347.295           0.0             No
1107       8325.27     832.527     9157.797           2.0             No
1108       8460.87     846.087     9306.957           16.0             Yes
1052       5792.09     579.209     6371.299           3.0             No

--- IT employees: 49 rows ---
      Bonus \
EmployeeID  Name  Age  Department  City  Gender  Salary  ExperienceYears  Rating  JoinDate
1119  Mohamed Hassan  33.0      IT  Ismailia  Male  4000.000000           10.0      4.0  2018-01-17
400.000000
1040  Salma Shafik  45.0      IT  Tanta  Female  10888.500000           5.0      3.0  2022-01-18
1088.850000
1095  Ahmed Brown  34.0      IT  Giza  Male  9394.746517           14.0      3.0  2014-08-09
939.474652
1039  Mona Davis  37.0      IT  Luxor  Male  12325.620000           15.0      3.0  2013-12-11
1232.562000
1036  Sophia Gaber  22.0      IT  Giza  Female  8221.620000           1.0      3.0  2012-07-05
822.162000
      TotalSalary  SeniorEmployee
```

EmployeeID												
1119	4400.000000		Yes									
1040	11977.350000		Yes									
1095	10334.221169		Yes									
1039	13558.182000		Yes									
1036	9043.782000		No									

--- Salary > 10000: 76 rows ---

	Name	Age	Department	City	Gender	Salary	ExperienceYears	Rating	JoinDate	Bo
nus	TotalSalary \									
EmployeeID										
1109	Dina Mostafa	33.0	Sales	Cairo	Male	13749.00	11.0	3.0	2023-10-27	1374.
900	15123.900									
1170	Nesma Wilson	24.0	Sales	Tanta	Male	17588.45	0.0	5.0	2023-02-17	1758.
845	19347.295									
1060	Mona Mostafa	33.0	Finance	Ismailia	Male	14092.25	13.0	4.0	2018-04-11	1409.
225	15501.475									
1032	Omar Miller	52.0	Sales	Mansoura	Male	14672.69	16.0	3.0	2018-08-14	1467.
269	16139.959									
1040	Salma Shafik	45.0	IT	Tanta	Female	10888.50	5.0	3.0	2022-01-18	1088.
850	11977.350									

SeniorEmployee

EmployeeID	
1109	Yes
1170	No
1060	Yes
1032	Yes
1040	Yes

--- Cairo & Rating>=4: 10 rows ---

	Name	Age	Department	City	Gender	Salary	ExperienceYears	Rating	JoinDate	Bon
us	TotalSalary \									
EmployeeID										
1052	David Abdallah	58.0	HR	Cairo	Male	5792.09	3.0	4.0	2016-11-06	579.2
09	6371.299									
1041	Heba Ramzy	37.0	Sales	Cairo	Male	13713.61	18.0	5.0	2012-01-06	1371.3
61	15084.971									
1144	Robert Saleh	50.0	Sales	Cairo	Male	7950.91	17.0	5.0	2011-04-28	795.0
91	8746.001									
1111	Linda Abdallah	28.0	Marketing	Cairo	Female	13748.22	14.0	5.0	2024-03-31	1374.8
22	15123.042									
1006	Sophia Nassar	24.0	HR	Cairo	Female	6494.08	8.0	4.0	2018-11-07	649.4
08	7143.488									

SeniorEmployee

EmployeeID	
1052	No
1041	Yes
1144	Yes
1111	Yes
1006	Yes

--- Sorted by Salary desc (top 5) ---

	Name	Age	Department	City	Gender	Salary	ExperienceYears	Rating	JoinDate	
Bonus	TotalSalary \									
EmployeeID										
1055	Patricia Shafik	22.0	Marketing	Mansoura	Male	18662.67	16.0	4.0	2023-01-13	18
66.267	20528.937									
1170	Nesma Wilson	24.0	Sales	Tanta	Male	17588.45	0.0	5.0	2023-02-17	17
58.845	19347.295									
1198	Ahmed Hassan	53.0	Finance	Luxor	Female	16059.11	11.0	5.0	2010-03-08	16
05.911	17665.021									
1029	Laila Fathy	26.0	Finance	Tanta	Male	14955.00	10.0	4.0	2014-06-13	14
95.500	16450.500									
1032	Omar Miller	52.0	Sales	Mansoura	Male	14672.69	16.0	3.0	2018-08-14	14
67.269	16139.959									

SeniorEmployee

EmployeeID	
1055	Yes
1170	No
1198	Yes
1029	Yes
1032	Yes

--- Sorted by JoinDate asc (top 5) ---

EmployeeID	Bonus	TotalSalary \	Name	Age	Department	City	Gender	Salary	ExperienceYears	Rating	JoinDate
1125			Rana Wilson	35.0	Finance	Alexandria	Male	13305.56	2.0	3.0	2010-01-10
	1330.556	14636.116									
1056			Mariam Wilson	54.0	IT	Mansoura	Male	8245.10	9.0	2.0	2010-01-12
	824.510	9069.610									
1071			Mariam Abdallah	47.0	Sales	Mansoura	Female	8825.67	14.0	2.0	2010-01-21
	882.567	9708.237									
1110			Dina Williams	30.0	IT	Cairo	Male	7741.97	17.0	2.0	2010-02-05
	774.197	8516.167									
1063			Mona Abdallah	56.0	Marketing	Giza	Male	8212.48	3.0	2.0	2010-02-12
	821.248	9033.728									

SeniorEmployee

EmployeeID

1125	No
1056	Yes
1071	Yes
1110	Yes
1063	No

--- Top 10 highest salaries ---

EmployeeID	Name	Department	Salary
1055	Patricia Shafik	Marketing	18662.67
1170	Nesma Wilson	Sales	17588.45
1198	Ahmed Hassan	Finance	16059.11
1029	Laila Fathy	Finance	14955.00
1032	Omar Miller	Sales	14672.69
1084	Michael Kamel	HR	14604.33
1060	Mona Mostafa	Finance	14092.25
1121	Linda Davis	HR	14054.61
1124	Yasmin Mostafa	Sales	13860.07
1177	Mariam Ramzy	HR	13823.96

Part 4: Aggregation

Code

```
section("PART 4: AGGREGATION")

# 25. Department-wise average salary
print("\n--- Average salary by department ---")
print(df.groupby("Department")["Salary"].mean().round(2))

# 26. Department-wise max/min salary
print("\n--- Max/Min salary by department ---")
print(df.groupby("Department")["Salary"].agg(["max", "min"]))

# 27. Count employees in each city
print("\n--- Employee count by city ---")
print(df.groupby("City").size())

# 28. Average rating by department
print("\n--- Average rating by department ---")
print(df.groupby("Department")["Rating"].mean().round(2))

# 29. Number of males and females
print("\n--- Gender counts ---")
print(df["Gender"].value_counts())

# 30. Average experience by department
print("\n--- Average experience by department ---")
print(df.groupby("Department")["ExperienceYears"].mean().round(2))

# 31. Highest paid employee in each department
print("\n--- Highest paid employee per department ---")
idx = df.groupby("Department")["Salary"].idxmax()
print(df.loc[idx, ["Name", "Department", "Salary"]])
```

Output

```
--- Average salary by department ---
Department
Finance      9592.27
HR            9734.52
IT            9268.73
Marketing     8774.07
Operations    9320.24
Sales         9361.91
Name: Salary, dtype: float64

--- Max/Min salary by department ---
           max      min
Department
Finance    16059.11  4000.00
HR         14604.33  4000.00
IT         13540.92  4000.00
Marketing  18662.67  4000.00
Operations 13226.57  5828.87
Sales      17588.45  4000.00

--- Employee count by city ---
City
Alexandria    27
Aswan         29
Cairo         24
Giza          21
Ismailia      22
Luxor         26
Mansoura      24
Tanta         27
dtype: int64

--- Average rating by department ---
Department
Finance      3.27
HR            3.00
IT            3.04
Marketing     3.24
```

Operations 3.13
Sales 2.86
Name: Rating, dtype: float64

--- Gender counts ---

Gender
Male 109
Female 91
Name: count, dtype: int64

--- Average experience by department ---

Department
Finance 9.61
HR 9.88
IT 10.10
Marketing 9.03
Operations 8.60
Sales 10.00
Name: ExperienceYears, dtype: float64

--- Highest paid employee per department ---

EmployeeID	Name	Department	Salary
1198	Ahmed Hassan	Finance	16059.11
1084	Michael Kamel	HR	14604.33
1114	Ahmed Aziz	IT	13540.92
1055	Patricia Shafik	Marketing	18662.67
1064	Michael Kamel	Operations	13226.57
1170	Nesma Wilson	Sales	17588.45

Part 5: Sqlite3

Code

```
section("PART 5: SQLITE3")

# 32. Create employee.db
conn = sqlite3.connect("employee.db")
cursor = conn.cursor()

# 33. Create employees table
cursor.execute("DROP TABLE IF EXISTS employees")
cursor.execute("""
CREATE TABLE employees (
    EmployeeID INTEGER PRIMARY KEY,
    Name TEXT,
    Age REAL,
    Department TEXT,
    City TEXT,
    Gender TEXT,
    Salary REAL,
    ExperienceYears REAL,
    Rating REAL,
    JoinDate TEXT,
    Bonus REAL,
    TotalSalary REAL,
    SeniorEmployee TEXT
)
""")
conn.commit()

# 34. Insert cleaned DataFrame using to_sql()
df_sql = df.reset_index() # bring EmployeeID back as a column
df_sql["JoinDate"] = df_sql["JoinDate"].dt.strftime("%Y-%m-%d")
df_sql.to_sql("employees", conn, if_exists="replace", index=False)

# 35. Read all records
all_records = pd.read_sql("SELECT * FROM employees", conn)
print(f"\n--- All records read back from SQLite: {len(all_records)} rows ---")

# 36. Display first 10 rows from SQLite
print("\n--- First 10 rows from SQLite ---")
print(pd.read_sql("SELECT * FROM employees LIMIT 10", conn))

# 37. Count total employees
total_count = pd.read_sql("SELECT COUNT(*) AS TotalEmployees FROM employees", conn)
print("\n--- Total employees ---")
print(total_count)

# 38. Average salary
avg_salary = pd.read_sql("SELECT AVG(Salary) AS AvgSalary FROM employees", conn)
print("\n--- Average salary ---")
print(avg_salary)

# 39. Maximum salary
max_salary = pd.read_sql("SELECT MAX(Salary) AS MaxSalary FROM employees", conn)
print("\n--- Maximum salary ---")
print(max_salary)

# 40. Employees with Salary > 10000
print("\n--- Employees with Salary > 10000 (first 10) ---")
print(pd.read_sql("SELECT * FROM employees WHERE Salary > 10000 LIMIT 10", conn))

# 41. Employees in IT
print("\n--- Employees in IT (first 10) ---")
print(pd.read_sql("SELECT * FROM employees WHERE Department = 'IT' LIMIT 10", conn))

# 42. Group by Department, AVG(Salary)
print("\n--- AVG(Salary) grouped by Department ---")
print(pd.read_sql(
    "SELECT Department, AVG(Salary) AS AvgSalary FROM employees GROUP BY Department", conn
))

# 43. Update Salary = Salary*1.05 where Department='HR'
cursor.execute("UPDATE employees SET Salary = Salary * 1.05 WHERE Department = 'HR'")
conn.commit()
```

```

print(f"\nUpdated HR salaries (+5%): {cursor.rowcount} rows affected")

# 44. Delete employees where Rating < 3.0
cursor.execute("DELETE FROM employees WHERE Rating < 3.0")
conn.commit()
print(f"Deleted employees with Rating < 3.0: {cursor.rowcount} rows affected")

# 45. Add a new employee using SQL INSERT
cursor.execute("""
INSERT INTO employees (EmployeeID, Name, Age, Department, City, Gender, Salary,
ExperienceYears, Rating, JoinDate, Bonus, TotalSalary, SeniorEmployee)
VALUES (9999, 'New Employee', 30, 'IT', 'Cairo', 'Male', 12000, 3, 4.5,
'2024-01-15', 1200, 13200, 'No')
""")
conn.commit()
print("\nInserted new employee with EmployeeID 9999")

# 46. Search employee by EmployeeID
print("\n--- Search EmployeeID = 9999 ---")
print(pd.read_sql("SELECT * FROM employees WHERE EmployeeID = 9999", conn))

# 47. Top 5 salaries
print("\n--- Top 5 salaries ---")
print(pd.read_sql("SELECT Name, Department, Salary FROM employees ORDER BY Salary DESC LIMIT 5", conn))

# 48. Order by JoinDate DESC
print("\n--- Ordered by JoinDate DESC (first 5) ---")
print(pd.read_sql("SELECT Name, JoinDate FROM employees ORDER BY JoinDate DESC LIMIT 5", conn))

# 49. Export final table back to CSV
final_table = pd.read_sql("SELECT * FROM employees", conn)
final_table.to_csv("employees_final.csv", index=False)
print(f"\nExported final table ({len(final_table)} rows) to employees_final.csv")

# 50. Close the database connection
conn.close()
print("\nDatabase connection closed.")

```

Output

--- All records read back from SQLite: 200 rows ---

--- First 10 rows from SQLite ---

	EmployeeID	Name	Age	Department	City	Gender	Salary	ExperienceYears	Rating	JoinDate
0	1109	Dina Mostafa	33.0	Sales	Cairo	Male	13749.00	11.0	3.0	2023-10-2
1	1170	Nesma Wilson	24.0	Sales	Tanta	Male	17588.45	0.0	5.0	2023-02-1
2	1107	Yasmin Mostafa	33.0	Finance	Luxor	Male	8325.27	2.0	3.0	2016-06-1
3	1108	Rana Fathy	40.0	Operations	Luxor	Female	8460.87	16.0	2.0	2011-03-2
4	1052	David Abdallah	58.0	HR	Cairo	Male	5792.09	3.0	4.0	2016-11-0
5	1034	John Fahmy	40.0	Marketing	Tanta	Female	4799.41	16.0	1.0	2023-02-2
6	1060	Mona Mostafa	33.0	Finance	Ismailia	Male	14092.25	13.0	4.0	2018-04-1
7	1153	Youssef Mostafa	40.0	Marketing	Giza	Male	9959.02	8.0	3.0	2012-09-1
8	1005	Aya Jones	50.0	Sales	Mansoura	Female	9808.30	15.0	3.0	2011-07-2
9	1032	Omar Miller	52.0	Sales	Mansoura	Male	14672.69	16.0	3.0	2018-08-1

	SeniorEmployee
0	Yes
1	No
2	No
3	Yes
4	No
5	Yes
6	Yes
7	Yes
8	Yes
9	Yes

--- Total employees ---

TotalEmployees

0 200

--- Average salary ---

AvgSalary

0 9348.346991

--- Maximum salary ---

MaxSalary

0 18662.67

--- Employees with Salary > 10000 (first 10) ---

	EmployeeID	Name	Age	Department	City	Gender	Salary	ExperienceYears	Rating	JoinDate	
	Bonus	TotalSalary \									
0	1109	Dina Mostafa	33.0	Sales	Cairo	Male	13749.00	11.0	3.0	2023-10-27	1
	374.900	15123.900									
1	1170	Nesma Wilson	24.0	Sales	Tanta	Male	17588.45	0.0	5.0	2023-02-17	1
	758.845	19347.295									
2	1060	Mona Mostafa	33.0	Finance	Ismailia	Male	14092.25	13.0	4.0	2018-04-11	1
	409.225	15501.475									
3	1032	Omar Miller	52.0	Sales	Mansoura	Male	14672.69	16.0	3.0	2018-08-14	1
	467.269	16139.959									
4	1040	Salma Shafik	45.0	IT	Tanta	Female	10888.50	5.0	3.0	2022-01-18	1
	088.850	11977.350									
5	1094	Linda Hassan	33.0	HR	Cairo	Female	11411.63	11.0	1.0	2010-04-16	1
	141.163	12552.793									
6	1113	Heba Nassar	40.0	HR	Luxor	Female	13717.50	15.0	4.0	2020-11-01	1
	371.750	15089.250									
7	1098	Linda Davis	56.0	Finance	Luxor	Male	12600.25	19.0	4.0	2017-03-21	1
	260.025	13860.275									
8	1152	David Gaber	28.0	Sales	Aswan	Female	11806.50	5.0	1.0	2016-10-02	1
	180.650	12987.150									
9	1039	Mona Davis	37.0	IT	Luxor	Male	12325.62	15.0	3.0	2013-12-11	1
	232.562	13558.182									

SeniorEmployee

0 Yes
1 No
2 Yes
3 Yes
4 Yes
5 Yes
6 Yes
7 Yes
8 Yes
9 Yes

--- Employees in IT (first 10) ---

	EmployeeID	Name	Age	Department	City	Gender	Salary	ExperienceYears	Rating	Joi
	nDate	Bonus \								
0	1119	Mohamed Hassan	33.0	IT	Ismailia	Male	4000.000000	10.0	4.0	2018-
	01-17	400.000000								
1	1040	Salma Shafik	45.0	IT	Tanta	Female	10888.500000	5.0	3.0	2022-
	01-18	1088.850000								
2	1095	Ahmed Brown	34.0	IT	Giza	Male	9394.746517	14.0	3.0	2014-
	08-09	939.474652								
3	1039	Mona Davis	37.0	IT	Luxor	Male	12325.620000	15.0	3.0	2013-
	12-11	1232.562000								
4	1036	Sophia Gaber	22.0	IT	Giza	Female	8221.620000	1.0	3.0	2012-
	07-05	822.162000								
5	1182	Farida Smith	58.0	IT	Alexandria	Female	6493.800000	17.0	2.0	2024-
	12-25	649.380000								
6	1012	Patricia Fahmy	40.0	IT	Giza	Female	11617.930000	5.0	4.0	2011-
	05-04	1161.793000								
7	1149	Dina Fathy	22.0	IT	Cairo	Male	8142.260000	10.0	3.0	2011-
	10-24	814.226000								
8	1017	Nour Ramzy	26.0	IT	Aswan	Female	7335.640000	18.0	4.0	2014-
	07-26	733.564000								
9	1049	Laila Nassar	56.0	IT	Alexandria	Male	7515.250000	19.0	4.0	2012-
	10-23	751.525000								

TotalSalary SeniorEmployee

0 4400.000000 Yes
1 11977.350000 Yes
2 10334.221169 Yes
3 13558.182000 Yes
4 9043.782000 No

5	7143.180000	Yes
6	12779.723000	Yes
7	8956.486000	Yes
8	8069.204000	Yes
9	8266.775000	Yes

--- AVG(Salary) grouped by Department ---

	Department	AvgSalary
0	Finance	9592.273425
1	HR	9734.518641
2	IT	9268.734951
3	Marketing	8774.073313
4	Operations	9320.238000
5	Sales	9361.906429

Updated HR salaries (+5%): 32 rows affected

Deleted employees with Rating < 3.0: 75 rows affected

Inserted new employee with EmployeeID 9999

--- Search EmployeeID = 9999 ---

	EmployeeID	Name	Age	Department	City	Gender	Salary	ExperienceYears	Rating	JoinDate	Bonus
0	9999	New Employee	30.0	IT	Cairo	Male	12000.0	3.0	4.5	2024-01-15	1200.0

	SeniorEmployee
0	No

--- Top 5 salaries ---

	Name	Department	Salary
0	Patricia Shafik	Marketing	18662.67
1	Nesma Wilson	Sales	17588.45
2	Ahmed Hassan	Finance	16059.11
3	Laila Fathy	Finance	14955.00
4	Omar Miller	Sales	14672.69

--- Ordered by JoinDate DESC (first 5) ---

	Name	JoinDate
0	Aya Ramzy	2024-12-06
1	Mohamed Williams	2024-12-03
2	Olivia Abdallah	2024-06-27
3	Rana Aziz	2024-05-10
4	Aya Mostafa	2024-04-05

Exported final table (126 rows) to employees_final.csv

Database connection closed.

Bonus Challenges

Code

Output

```
[load_data] Loaded 215 rows from 'EDA_SQLite_Employee_200.csv'  
[clean_data] Cleaned dataset now has 200 rows and 0 missing values  
[save_to_sqlite] Saved 200 rows to 'employee_bonus.db' (table: employees)  
[load_data] ERROR: File not found -> this_file_does_not_exist.csv
```

All 50 tasks + bonus challenges completed successfully.